

Surabhi Vishwasrao

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EDUCATION

Sardar Patel Institute Of Technology <i>B.Tech in Electronics and Telecommunication</i> Relevant Coursework: Cybersecurity and Digital Forensics, Computer Communication Networks, Object Oriented Programming	Mumbai 2022 – 2026
Sardar Patel Institute Of Technology <i>Minors in Computer Engineering</i> Relevant Coursework: Data Structures and Algorithms, Database Management System, Operating System	Mumbai 2022 – 2026
PACE Junior College <i>Grade: 90.71%</i> Relevant Coursework: Communication Skills, Analytical and Critical Thinking	Mumbai 2020 – 2022
Universal High School <i>Grade: 97.50%</i>	Thane 2010 – 2020

TECHNICAL SKILLS

Languages: Java, C++, JavaScript (basic), Python
Databases: MySQL
Tools & Libraries: PyAudioAnalysis, pandas, SpeechRecognition, logging
Technologies: Git, GitHub, APIs, Responsive Web Design
DevOps: Docker
Operating System: Linux, Unix, Windows

SOFTWARE PROJECTS

- Multispeaker Audio Analytics System using Python**
- Built a Python system for transcribing and segmenting multi-speaker audio using clustering and STT.
 - Used PyAudioAnalysis, pandas, and SpeechRecognition for audio processing and diarization.
 - Demonstrated problem-solving and ownership in handling end-to-end design and testing.
- AI-Powered Travel Itinerary Generator** (*MongoDB, Express.js, React.js, Node.js, Google Maps API, OpenAI/Gemini API, JWT, Tailwind CSS*)
- Generated personalized, AI-driven travel itineraries based on destination, duration, budget, and user preferences.
 - Integrated Google Maps API for interactive maps, points of interest, and optimized route suggestions.
 - Implemented secure JWT-based authentication for itinerary saving, editing, and sharing.
 - Designed a responsive and mobile-friendly interface using Tailwind CSS.
 - Built a scalable backend with Express.js and MongoDB for efficient itinerary storage and retrieval.
- Marsupial Robotic System for Vision-based Terrain Exploration** (*Ongoing*)
- Developing an intelligent robotic platform using UGV02, Raspberry Pi, and LiDAR for real-time environment sensing.
 - Integrating camera-based vision processing and path planning for semi-autonomous terrain navigation.

POSITIONS OF RESPONSIBILITY

- Teaching Assistant – Digital Signal Processing (DSP)**
- Mentored students, simplified DSP concepts, and created structured notes to support academic performance.
- Mentor – Abhyudaya (Education Support Program - SPJIMR)**
- Demonstrated empathy, cultural awareness, and strong interpersonal communication while mentoring underprivileged school children.

PROFESSIONAL PUBLICATIONS

- A Detailed Overview of Brain-Machine and Brain-Computer Interfaces**
International Journal of Innovative Science and Research Technology
- The paper provides a comprehensive overview of brain-computer and brain-machine interfaces (BCIs/BMIs), focusing on their working principles, signal acquisition methods (like EEG), and data processing techniques.
 - Link: <https://www.ijisrt.com/a-detailed-overview-of-braincomputer-and-brainmachine-interfaces>